Verified Stateflow v85 integration with a 500 m³ pool

Actually executed MATLAB/Simulink R2026a | new v03 model | 1 hour

1. Architecture, provenance and boundary

The original v85 model is re-executed without modifications. Its certified direct heat pump service is redirected partly to the pool and the remainder continues to the existing district-heat customer. This is NOT additional waste heat recovery. No 8760-hour v85 record is available. V03 is a one-way output-trace link, not a closed-loop dispatcher.

$$E_{pool} + E_{DH,remaining} = E_{HP,service,v85}$$

2. Heat allocation and trace reconstruction

The input signal is reconstructed from the valid 80-component macrosegments in the 1380-column v85 trace. Direct HP service is z(25), in MW. Segment duration uncertainty lies in z(7:8), in hours. Integrated service must fall inside the original independently reported certified energy hull.

$$E_k\,[\mathrm{MWh}] = \sum_s z25(s) \cdot \frac{\tau_{low}(s) + \tau_{high}(s)}{2}$$

$$Q_k\,[\mathrm{W}] = 10^6 \cdot E_k\,[\mathrm{MWh}] / dt_k\,[\mathrm{h}]$$

3. Pool thermostat and water energy balance

The 500 m³ pool has thermal capacity C = 2.084727 GJ/K. Model losses include evaporation, convection, radiation, construction and make-up water. Hypothetical parameters are not measurements of the actual site.

$$C = \rho \cdot V \cdot c_p = 997 \cdot 500 \cdot 4182 = 2\,084\,727\,000\ \mathrm{J/K}$$

$$E(16 \rightarrow 28\ ^\circ\mathrm{C}) = 12 \cdot C / 3.6e9 = 6.94909\ \mathrm{MWh}\ (\text{no losses})$$

$$T(k+1) = T(k) + dt\,[\mathrm{s}] \cdot (Q_{pool} - Q_{loss}) / C$$

$$Q_{safe} = \max(0,\ C \cdot (27.75 - T(k)) / 900 + Q_{loss})$$

$$Q_{pool} \le \min(Q_{v85},\ 150\,000\ \mathrm{W},\ Q_{safe})$$

$$Q_{DH,remaining} = Q_{v85} - Q_{pool}$$

4. Executed scenarios and measured simulation outputs

These three cases use the bitwise-verified original service_E3.0_a1.0_H2_dt0.25 trace. The further charge_E4.95_a1.0_H2_dt0.25 and binding_stop traces contain no direct service suitable for pooling. Nine Simulink scenarios executed in all.

Condition	Pool [MWh]	DH [MWh]
Cold water 16 °C	0.150000	0.297566
Warm water 27.70 °C	0.000000	0.447566
Pool STOP	0.000000	0.447566

0.447565789 MWh = 0.150000000 MWh + 0.297565789 MWh

5. Independent QA and remaining work

99 independent checks passed: original v85 SHA-256, CSV length, allocation and water first-law balances, pool STOP, temperature bound and no redirection of TES charging. The newly run v85 outputs match the previous MAT evidence bit for bit.

V03 is not closed-loop Stateflow control, extra heat recovery or a yearly data-centre simulation. Vendor COP maps, real hydraulic sizing, instrument validation and a 8760-hour measured time series are still absent.

SHA256 v85: 2a9de22c0c55e8c4a87f7b0e0cfe9cd97db08c991f09d98361b009de310adde7

Sources: v85 trace, saved MAT v86, 9 CSV, independent QA report.

EU 2024/1364: https://eur-lex.europa.eu/eli/reg_del/2024/1364/oj

STN EN 15288-1+A1:2024: https://normy.normoff.gov.sk/norma/139372/nahlad/

Slov-Lex 308/2012 published historical text: https://static.slov-lex.sk/static/SK/ZZ/2012/308/vyhlasene_znenie.print.html

6. Charts of executed cases

Zdroj: Simulink v03, 4 intervaly po 15 minut; popisky grafov v slovencine.



